ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 2ND YEAR

ANALOGUE ELECTRONICS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BEng Mechatronics and Robotics Engineering, 2nd year final examination: diodes and power supplies, BJT biasing and small-signal amplifiers, operational amplifier circuits and their limitations, amplifier concepts and feedback, oscillators, sampling and data converters.

Written by Academic Central for practice, following analogue electronics modules at South African universities (e.g. University of Pretoria ENE 310 Analogue Electronics: amplifier concepts, op-amp circuits, oscillators and data conversion). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Show ALL your working, including formulae and substitutions. Give answers correct to TWO decimal places, with units.
3. Unless stated otherwise, use V_D = 0.7 V for a conducting silicon diode, V_BE = 0.7 V, and V_T = 25 mV. Op-amps are ideal unless a limitation is given.
4. Non-programmable scientific calculators may be used.
5. Number your answers exactly as the questions are numbered.

QUESTION 1: DIODES AND POWER SUPPLIES

1.1 A transformer with a 12 V RMS secondary feeds a full-wave bridge rectifier, a 2 200 µF smoothing capacitor and a load that draws 100 mA. The mains frequency is 50 Hz.

1.1.1 Calculate the peak voltage across the capacitor. (3)

1.1.2 Calculate the peak-to-peak ripple voltage. (3)

1.1.3 Estimate the average DC output voltage. (2)

1.1.4 Calculate the peak inverse voltage (PIV) that each diode must withstand. (2)

1.2 A 5.1 V Zener diode regulator is fed from 15 V through a 220 Ω series resistor and supplies a 510 Ω load. Calculate the series current, the load current, the Zener current and the power dissipated in the Zener diode. (6)

1.3 Sketch the output waveform of a half-wave rectifier without a capacitor, and give TWO reasons why a full-wave bridge is preferred in a robot's power supply. (4)

[20]

QUESTION 2: BJT AMPLIFIER

FIGURE 1 shows a voltage-divider biased common-emitter amplifier. Use the approximate method (assume the base current is negligible compared with the current in R1 and R2).

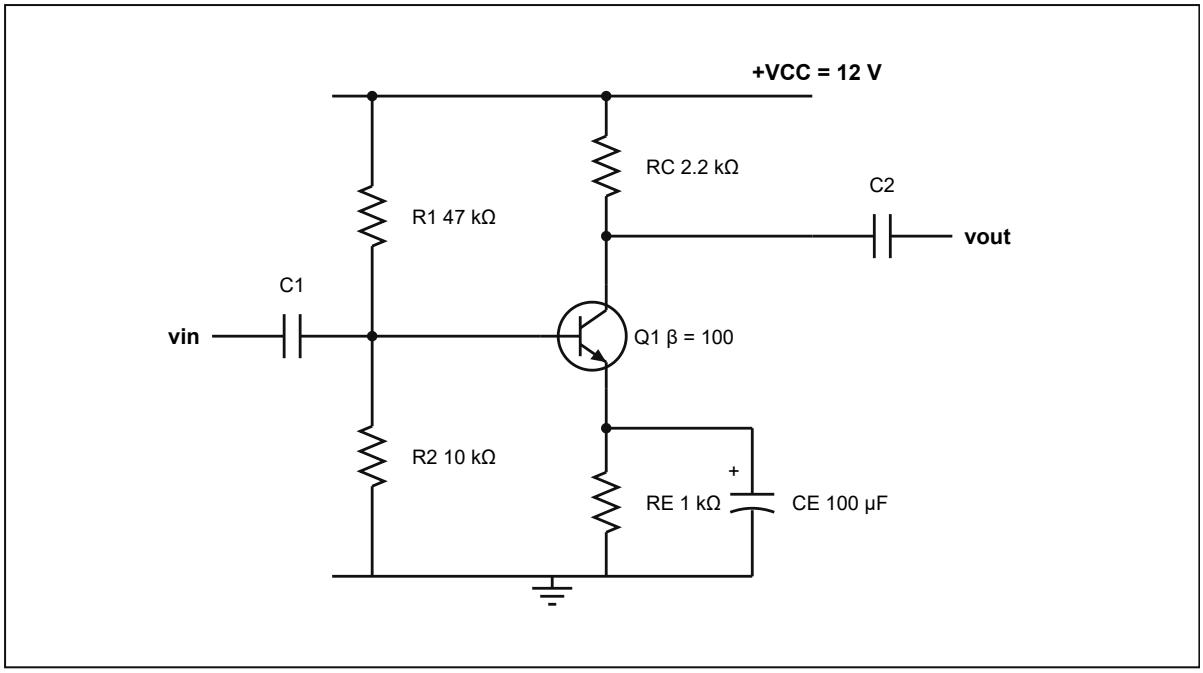


FIGURE 1: Common-emitter amplifier

2.1 DC analysis:

2.1.1 Calculate the base voltage V_B. (2)

2.1.2 Calculate the emitter voltage V_E and the emitter current I_E. (2)

2.1.3 Write down the collector current I_C. (1)

2.1.4 Calculate V_C and V_CE. (3)

2.1.5 Is the transistor in the active region? Give a reason. (1)

2.2 Small-signal analysis:

2.2.1 Calculate the AC emitter resistance r_e and the voltage gain with RE fully bypassed by CE. (3)

2.2.2 Calculate the voltage gain if CE is removed. (3)

2.2.3 Explain the trade-off between the two gains in terms of stability. (2)

2.3 Calculate the input impedance of the amplifier with CE connected. (3)

2.4 Calculate the two end points of the DC load line, I_C(sat) and V_CE(cut-off). (3)

2.5 What is the purpose of the coupling capacitors C1 and C2? (2)

[25]

QUESTION 3: OPERATIONAL AMPLIFIER CIRCUITS

3.1 A non-inverting amplifier has Rf = 47 kΩ and R1 = 4.7 kΩ. Calculate the gain and the output voltage for an input of 0.3 V. (3)

3.2 An inverting summing amplifier has Rf = 10 kΩ and three 10 kΩ input resistors. The inputs are 1 V, 2 V and −0.5 V. Calculate the output voltage. (3)

3.3 A difference amplifier has R1 = R3 = 10 kΩ and R2 = R4 = 100 kΩ, so that V_out = (R2/R1)(V_2 − V_1). A strain-gauge bridge gives V_2 = 2.05 V and V_1 = 2.00 V.

3.3.1 Calculate the output voltage. (3)

3.3.2 Explain what the common-mode rejection ratio (CMRR) is and why it matters for this sensor. (2)

3.4 The op-amp used in 3.1 has a gain-bandwidth product of 1 MHz. Calculate the bandwidth of the amplifier. (2)

3.5 An op-amp has a slew rate of 0.5 V/µs. Calculate the highest frequency at which it can deliver a 10 V peak sine wave without distortion. (3)

3.6 A first-order RC low-pass filter has R = 10 kΩ and C = 15 nF.

3.6.1 Calculate the cut-off frequency. (2)

3.6.2 What is the gain in dB at the cut-off frequency, and roughly at ten times the cut-off frequency? (2)

3.7 An inverting Schmitt trigger uses an op-amp that saturates at ±12 V, with R1 = 10 kΩ from the non-inverting input to ground and R2 = 40 kΩ from the output to the non-inverting input.

3.7.1 Calculate the upper and lower threshold voltages and the hysteresis. (3)

3.7.2 Why is a Schmitt trigger used to read a noisy sensor signal? (2)

[25]

QUESTION 4: AMPLIFIER CONCEPTS AND FEEDBACK

4.1 Two amplifier stages with gains of 20 dB and 30 dB are cascaded. Calculate the overall gain in dB and as a ratio. (3)

4.2 An amplifier has an open-circuit voltage gain of 50, an input resistance of 2 kΩ and an output resistance of 100 Ω. It is driven by a 10 mV source with a 500 Ω internal resistance and drives a 400 Ω load. Calculate the output voltage and the overall voltage gain. (4)

4.3 An amplifier with open-loop gain A = 10 000 uses negative feedback with β = 0.01.

4.3.1 Calculate the closed-loop gain. (2)

4.3.2 Calculate the closed-loop gain if A falls by 50%, and comment on the result. (2)

4.4 State the maximum theoretical efficiency of a class A and a class B power amplifier, and define total harmonic distortion (THD). (4)

[15]

QUESTION 5: OSCILLATORS AND DATA CONVERSION

5.1 State the Barkhausen criterion for oscillation. (2)

5.2 A Wien-bridge oscillator uses R = 10 kΩ and C = 10 nF. Calculate the frequency of oscillation and state the gain the amplifier must have. (3)

5.3 A 10-bit ADC has a reference voltage of 5 V.

5.3.1 Calculate its resolution. (2)

5.3.2 Calculate the digital output, in decimal and in binary, for an input of 3.2 V. (3)

5.4 A vibration sensor produces signals up to 4 kHz. State the minimum sampling frequency and explain what aliasing is. (3)

5.5 A 4-bit R-2R DAC has a reference of 8 V and a full-scale output of Vref × (code/16). Calculate the output for the input 1010. (2)

[15]

TOTAL: 100

END OF QUESTION PAPER